

Cloud Computing Practical Examination

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Total Marks: 40

Duration: 60 Minutes Instructions

- Perform all tasks on your own cloud virtual machine.
 - You may use AWS EC2, GCP Compute Engine VM, or Azure Virtual Machine.
 - Attach screenshots as evidence of completion.
 - Create a single PDF containing all screenshots and answers.
 - Mention question numbers clearly.
 - Ensure all screenshots clearly show successful execution of commands.
 - Look at the end of the pdf for submission instructions
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Question 1: Docker Installation & Nginx Deployment

Task

1. Install Docker on your cloud virtual machine.
2. Verify Docker installation.
3. Download the Nginx image.
4. Run the Nginx container.
5. Access the Nginx welcome page from your browser.

Required Screenshots

- Docker version output • Docker images output
- Docker ps output showing Nginx running
- Browser showing "Welcome to nginx!"

Hint

Relevant Docker topics:

- Images
- Containers
- Port Mapping
- Container Lifecycle

```
Setting up dnsmasq-base (2.92-1ubuntu0.3) ...
Setting up runc (1.4.0-0ubuntu1) ...
Setting up dns-root-data (2025080400build1) ...
Setting up bridge-utils (1.7.1-4ubuntu3) ...
Setting up containerd (2.2.2-0ubuntu1) ...
Setting up ubuntu-fan (0.12.17) ...
Created symlink '/etc/systemd/system/multi-user.target.wants/ubuntu-fan.service' -> '/usr/lib/systemd/system/ubuntu-fan.service'.
Setting up docker.io (29.1.3-0ubuntu4.1) ...
Processing triggers for dbus (1.16.2-2ubuntu4) ...
Processing triggers for man-db (2.13.1-1build1) ...
Scanning processes...
Scanning linux images...

Running kernel seems to be up-to-date.

No services need to be restarted.

No containers need to be restarted.

No user sessions are running outdated binaries.

No VM guests are running outdated hypervisor (qemu) binaries on this host.
ubuntu@ip-172-31-24-100:~$ sudo systemctl start docker
sudo systemctl enable docker
Synchronizing state of docker.service with SysV service script with /usr/lib/systemd/systemd-sysv-install.
Executing: /usr/lib/systemd/systemd-sysv-install enable docker
ubuntu@ip-172-31-24-100:~$ docker --version
Docker version 29.1.3, build 29.1.3-0ubuntu4.1
ubuntu@ip-172-31-24-100:~$
```

Question 2: Docker Volume Practical

Task

1. Create a Docker volume named: studentdata
2. Run an Ubuntu container using the volume.
3. Create a file: /data/info.txt
4. Write your name and roll number inside the file.
5. Display the contents of the file.

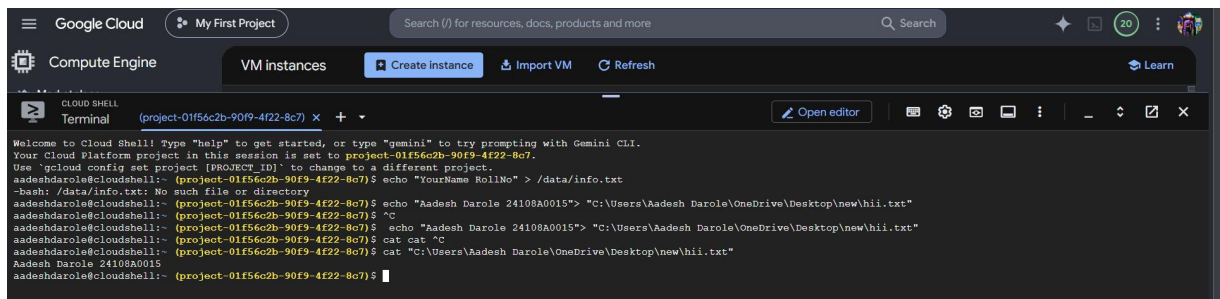
Required Screenshots

- Volume creation
- Running container
- File creation
- File contents displayed using cat

```
Google Cloud My First Project Search (V) for resources, docs, products and more
Compute Engine VM instances Create instance Import VM Refresh Learn
CLOUD SHELL Terminal (project-01f56c2b-90f9-4f22-8c7) x + Open editor
6376489be516: Pull complete
57b3fbf43092: Pull complete
d4dcd8aeeed: Pull complete
41c9f6f90940: Pull complete
c5a7565de4cf: Pull complete
80fa08b690ad: Pull complete
72c03230f136: Pull complete

Digest: sha256:608a100c71651bf5b773c89083b4alad7ef4b2bd05d7a7e552271e03123692ad
Status: Downloaded newer image for nginx:latest
docker.io/library/nginx:latest
aadeshdarole@cloudshell:~ (project-01f56c2b-90f9-4f22-8c7)$ sudo docker ps
CONTAINER ID   IMAGE      COMMAND                  CREATED    STATUS    PORTS    NAMES
9022ba36938c   nginx     "/docker-entrypoint.…"   2 minutes ago    Up 2 minutes    0.0.0.0:80->80/tcp    nginx-container
aadeshdarole@cloudshell:~ (project-01f56c2b-90f9-4f22-8c7)$ sudo docker images

IMAGE          ID              DISK USAGE  CONTENT SIZE  EXTRA
nginx:latest   608a100c7165   241MB      60MB          [REDACTED]
aadeshdarole@cloudshell:~ (project-01f56c2b-90f9-4f22-8c7)$ sudo docker volume create studentdata
studentdata
aadeshdarole@cloudshell:~ (project-01f56c2b-90f9-4f22-8c7)$ sudo docker volume ls
DRIVER      VOLUME NAME
local      studentdata
aadeshdarole@cloudshell:~ (project-01f56c2b-90f9-4f22-8c7)$ sudo docker run -it --name ubuntu-volume \
-v studentdata:/data ubuntu bash
Unable to find image 'ubuntu:latest' locally
latest: Pulling from library/ubuntu
1c24335ddd46: Pull complete
6f5c5aa4e145: Pull complete
9bc1f40d7f06: Download complete
Digest: sha256:f3d28607ddd78734bb7f71f17f3c6706c666b8b76cbff7c9ff6e5718d46ff64
Status: Downloaded newer image for ubuntu:latest
root@acb78910e2d8:/#
```



Question 3: Build Your First Docker Image

Task

Create a custom Docker image that displays the following message when executed:

Cloud Computing Practical Exam Completed Successfully

Build the image and run it successfully.

Required Screenshots

- Dockerfile
- docker build output
- docker images output
- docker run output

Hint

Relevant Dockerfile keywords:

- FROM
- CMD
- COPY

Think about:

- Which base image should be used?
- How does Docker know what command to execute when the container starts?
- When would COPY be useful while building an image?

```
GNU nano 8.7.1 Dockerfile *
FROM alpine:latest
CMD ["echo", "Cloud Computing Practical Exam Completed Successfully"]

Write to File: Dockerfile
^G Help      M-D DOS Format  M-A Append     M-B Backup File  ^M Browse
^C Cancel    M-M Mac Format  M-F Prepend     ^O Discard buffer
```

```
aws  [Icons] Search [Alt+S] United States (N. Virginia) Vikas-vm (019711414011) Vikas-vm

ubuntu@ip-172-31-24-100:~$ mkdir custom-build && cd custom-build
ubuntu@ip-172-31-24-100:~/custom-build$ nano Dockerfile
ubuntu@ip-172-31-24-100:~/custom-build$ sudo docker build -t my-custom-exam .
[+] Building 1.3s (5/5) FINISHED
=> [internal] load build definition from Dockerfile
=> => transferring dockerfile: 127B
=> [internal] load metadata for docker.io/library/alpine:latest
=> [internal] load .dockerignore
=> => transferring context: 2B
=> [1/1] FROM docker.io/library/alpine:latest@sha256:a2d49ea686c2adfe3c992e47dc3b5e7fa6e6b5055609400dc2acaeb241c829f4
=> => resolve docker.io/library/alpine:latest@sha256:a2d49ea686c2adfe3c992e47dc3b5e7fa6e6b5055609400dc2acaeb241c829f4
=> => sha256:9b70e313681f44d32991ec943f89228bc91d7431d4a84feafc269a76e3f96a63 3.87MB / 3.87MB
=> exporting to image
=> => exporting layers
=> => exporting manifest sha256:d4e3016bb6b76c36d4dcd3457f4398dee300681ca3c2a26f3f34f96f4922ac6
=> => exporting config sha256:239db0bbba399ace0fcb50dc7804691dbba80b5450b2ee820b75e1c2b4f1e31
=> => exporting attestation manifest sha256:3d6a30e718aea6ac52ec393bca95ff24620478ea9026a5f5c4b96b633199d71e
=> => exporting manifest list sha256:4bcee524a7c31ddba55614d6d70a00e8abc6af9115e0416de4c9a2cd65f98de6
=> => naming to docker.io/library/my-custom-exam:latest
=> => unpacking to docker.io/library/my-custom-exam:latest
ubuntu@ip-172-31-24-100:~/custom-build$

i-0642314b3d4e32627 (vikas-vm)
PublicIPs: 98.94.90.172 PrivateIPs: 172.31.24.100
```

Bonus Challenge (Optional)

Explain in 2–3 lines each:

1. Difference between a Docker Image and a Docker Container.
2. Why are Docker Volumes used?

Submission Checklist

- ☐ Docker Installation Verification
- ☐ Nginx Running Successfully
- ☐ Browser Access to Nginx
- ☐ Docker Volume Creation

- ☐ File Creation and Verification
 - ☐ Custom Docker Image Build
 - ☐ Custom Docker Container Execution
 - ☐ Bonus Questions (Optional)
-

Submission Instructions

After completing the practical examination:

Step 1: Create a PDF

Combine all screenshots and answers into a single PDF document.

Naming Convention

studentname_id_branch.pdf Example:

ImranSayyed_12345_CSE.pdf

Step 2: Repository Information

Repository:

<https://github.com/sayyedimran2599/azure-services>

Submission Branch:

practical_exam

Step 3: Switch to Submission Branch

If the branch is not already available locally, fetch the latest branch information from GitHub.

Switch to the branch:

practical_exam

Step 4: Copy Your PDF

Place the completed PDF inside the repository folder.

Verify that the PDF is visible inside the repository.

Step 5: Commit Message Format

Use the following format:

Name_ID_Q1Q2Q3

Examples:

ImranSayyed_12345_Q1Q2Q3

AishwaryaYadav_67890_Q1Q2

RahulPatil_54321_Q1

Where:

Q1Q2Q3 = All questions attempted

Q1Q2 = Questions 1 and 2 attempted

Q1 = Only Question 1 attempted

Step 6: Push Your Submission

Push your changes to the:

“practical_exam” branch.

Step 7: Verify Submission

Verify that:

- Your PDF is present in the repository.
- Your commit message follows the required format.
- The file is visible in the practical_exam branch.

Your submission is considered complete only after the PDF is successfully pushed to GitHub.

